

It can be calculated statistically using the formula below:

$$|\ln(n)/\ln(P)|$$

n = total number of bets

ln = natural logarithm

P = Probability

|.| = absolute value or 'modulus'

To make our analysis easy, I will provide the details. For 100 bets, with a 50% win rate, there is a possibility of a 9 times expected losing streak.

Coming back to our John and Tom example, either one can get 9 successive losing trades at any point in time.

John decides to risk 2% of his capital for every trade, and Tom risks 5% of his capital for every trade.

Let's see their account size after facing the 9 successive failed trades (assuming both don't deviate from the plan).

Trade #	Sequence	John's Risk (per trade)	John's Capital Value	Tom's Risk (per trade)	Tom's Capital Value
1	Loss	2%	98000	5%	95000
2	Loss	2%	96000	5%	90000
3	Loss	2%	94000	5%	85000
4	Loss	2%	92000	5%	80000
5	Loss	2%	90000	5%	75000
6	Loss	2%	88000	5%	70000
7	Loss	2%	86000	5%	65000
8	Loss	2%	84000	5%	60000
9	Loss	2%	82000	5%	55000



After 9 successive failed trades, John is left with a capital of 82,000, and he can still maintain his composure and take trades effectively.

Whereas Tom is left with 55,000 which is 45% of erosion on his account. So, he will face much emotional trauma to take further trades.